

2024 Spring VLSI DSP Homework Assignment #2

Due date: 2024/3/31

Q1. LMS filter design

For a **least mean square (LMS) adaptive filter**, assume the filter is of the form finite impulse response (FIR) and m -tap long (i.e., with m coefficients $b_0 \sim b_{m-1}$ for $x(n) \sim x(n-m+1)$). Given an input signal consisting of 2 frequency components

$$s(n) = \sin(2\pi * n / 16) + 0.5 * \cos(2 \pi * n / 4)$$

develop an adaptive low pass filter design

Set the target as a low pass filter to remove the high frequency component $\cos(2 \pi * n / 4)$ and use $\sin(2\pi * n / 16)$ as the desired (or training) signal for LMS adaptation.

- write a Matlab code to simulate the LMS based adaptive filtering. Calculate the **RMS** (root mean square) value of the latest 16 prediction errors (i.e., $r(n) = \text{sqrt}((e^2(n) + e^2(n-1) + \dots + e^2(n-15))/16)$ and the adaptation is converged if this value is less than 5% of RMS (root mean square) value of the desired signal, which equals $0.05/\text{sqrt}(2)$.
- Please determine the minimum filter length m and the step size μ so that the adaptation can converge in no more than 5000 iterations. The initial values of the filter coefficients $b_0 \sim b_{m-1}$ are set to $1/m$.
- Show the plot of " $r(n)$ " versus " n " and indicate when the filter converges, i.e. how many training samples are required
- Show the plot of filter coefficients $b_i(n)$, for $i = 0 \sim m-1$, versus " n " and see if the values of filter coefficients remain mostly unchanged after convergence
- Apply a 64-point FFT to the impulse response of the converged filter and verify the filter is indeed a low pass one. Note that the input vector to the 64-point FFT is $(b_0, b_1, \dots, b_{m-1}, 0, 0, \dots, 0)$ with $64-m$ trailing zeros.
- Conduct simulation with a sufficiently large number of samples to see how small the value of " r " can be (the convergence bias)

Q2. Discrete Wavelet Transform

For a discrete wavelet transform (DWT) adopting (9/7) filters, i.e. the low pass filter $h(i)$ is 9-taped and the high pass filter $g(i)$ is 7-taped. Both filters are liner phased and have symmetric coefficients. The filter coefficients are given in Table 1. For a corresponding inverse discrete wavelet transform, the low pass filter $q(i)$ is 7-taped and the high pass filter $p(i)$ is 9-taped. The filter coefficients are given in Table 2.

Table 1. Analysis filter coefficients for the floating point 9/7 filter

Analysis Filter Coefficients		
i	Lowpass Filter h_i	Highpass Filter g_i
0	0.852698679009	-0.788485616406
± 1	0.377402855613	0.418092273222
± 2	-0.110624404418	0.040689417609
± 3	-0.023849465020	-0.064538882629
± 4	0.037828455507	

Note: the high pass and low pass filter notations here are opposite to those in the lecture note

Table 2. Synthesis filter coefficients for the floating point 9/7 filter

Synthesis Filter Coefficients		
i	Low pass Filter q_i	High pass Filter p_i
0	0.788485616406	-0.852698679009
± 1	0.418092273222	0.377402855613
± 2	-0.040689417609	0.110624404418
± 3	-0.064538882629	-0.023849465020
± 4		-0.037828455507

- a) For a 512×512 gray scale image (will be provided along with the homework assignment), please conduct a 2-D 3-level DWT transform (as shown in Figure 1) and show the transformed result. Then conduct a 2-D 3-level IDWT to convert it back. Please compare if the reconstructed image (after IDWT) is same as the original image by calculating its PSNR value.



- b) By setting all three level 1 sub-bands HL1, LH1 and HH1 coefficients to zeros and perform IDWT. See how the reconstructed image is different from the original one and calculate its PSNR value.

Note 1: the filter orders for analysis (DWT) is (low pass 9/ high pass 7) and the filter orders for synthesis is (low pass 7/ high pass 9). And the filter coefficients have the following relations

$$h(i) = (-1)^{i+1} p(i), \quad g(i) = (-1)^i q(i)$$

Note 2: when performing down sampling in each octave, low pass filters always keep the **odd** numbered output data while high pass filters always keep the **even** numbered output data. In up-sampling process of IDWT, the discarded data are replaced with zeros and inserted to the output data stream.

Note 3: at the boundary of the image, you need to perform a symmetric extension to obtain the pixel values for $h(i)$, $g(i)$, $i < 0$. The extension is illustrated in Figure 3.

Note 4: The PSNR is calculated as

$$MSE = \frac{1}{M \cdot N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \left(I(i, j) - \hat{I}(i, j) \right)^2$$

$$PSNR = 10 \log_{10} \left(\frac{MAXI^2}{MSE} \right)$$

Where $I(i, j)$ is the original image and $\hat{I}(i, j)$ is the reconstructed image after performing DWT and IDWT. MAXI is the maximum possible value of a pixel. If it's an 8-bit pixel, MAXI is 255. For a reconstructed image with a PSNR value greater than 50dB, it is considered visually lossless.

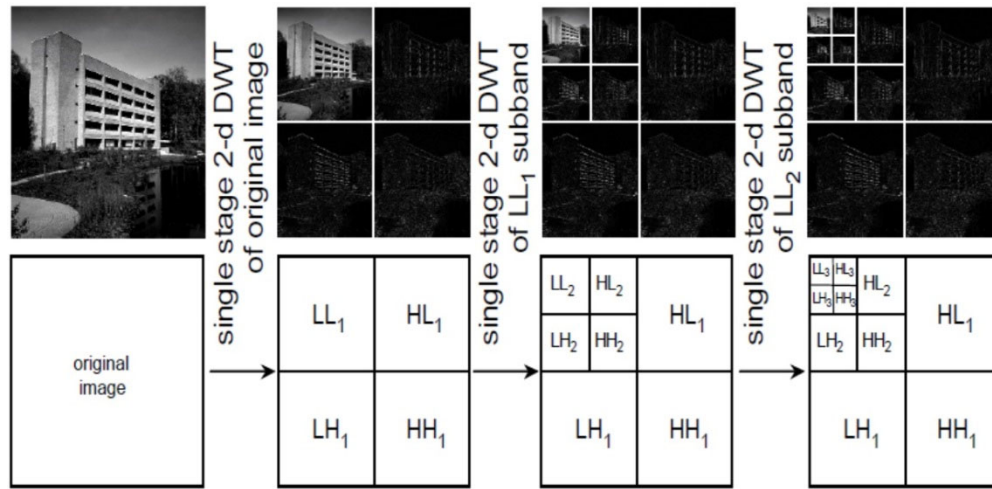
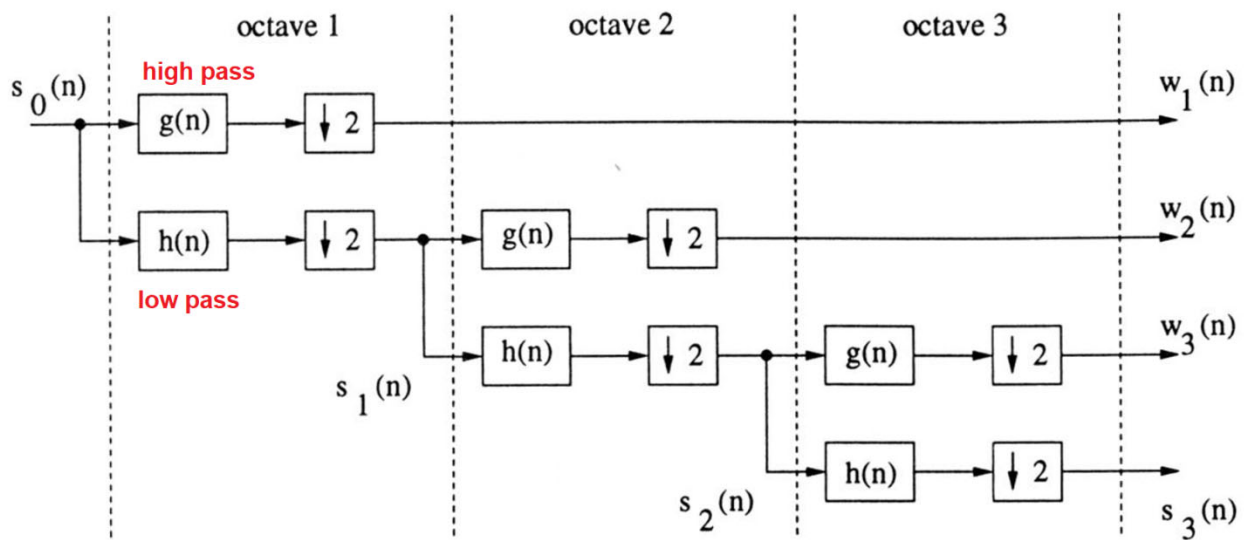
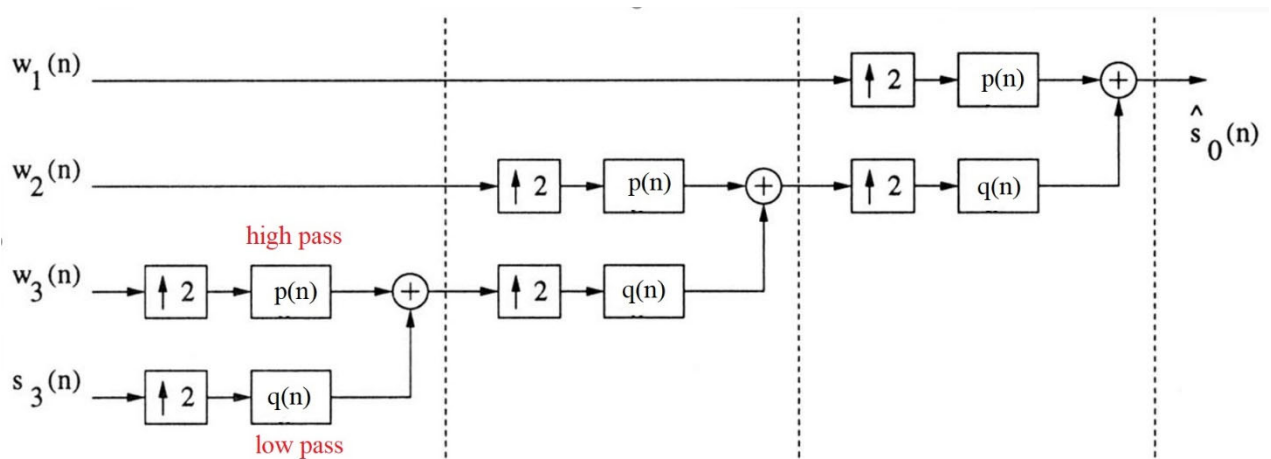


Figure 1. a 3-level DWT example



(a) One-dimensional 3-level DWT transform



(b) One-dimensional 3-level IDWT transform

Figure 2. 1-D 3-level DWT versus IDWT



Figure 3. Symmetric extension scheme for boundary pixels